

FMIS.Agro - Production Planning ERP

Overview: The most critical agricultural process: planning & discipline. FMIS.Agro guides farmers through a structured 90-day plan. 6-layer ERP architecture.

Features:

- Demand Planning: Yield forecasting, market analysis
- Master Production Schedule (90-day weekly)
- Material Requirements Planning (seed, fertilizer, pesticide timing)
- Capacity Planning (labor & equipment allocation)
- Cost Accounting (budget tracking)
- Daily Companion: Discipline enforcement

Expected Outcomes:

- ✓ +20% Yield (40 ton/ha → 48 ton/ha)
- ✓ -15% Costs (optimal input utilization)
- ✓ 96% Plan Adherence (vs. 70% intuition-based)
- ✓ +15-20% Income (1.1M → 1.3M TL/season)

FMIS.Monitor - Satellite + IoT Monitoring

Overview: Continuous crop health surveillance. Satellite (NDVI), IoT sensors, AI disease detection. Farmer alerted: stress 10-14 days early, disease 72 hours early.

Features:

- Satellite NDVI: Crop health index daily
- IoT Sensors: Soil, moisture, temperature
- AI Disease Detection: 94% accuracy on Turkish crops
- Weather API: 10-day forecast, risk alerts

Expected Outcomes:

- ✓ 90% Disease Prevention
- ✓ -20% Pesticide Usage
- ✓ Yield Stability: $\pm 25\%$ $\rightarrow$ $\pm 10\%$
- ✓ +12% Medicine Cost Savings

FMIS.Livestock - Veterinary AI

Overview: For livestock operations: wearable technology (ear tags, neck collars) + AI veterinary diagnostics. Detects mastitis, lameness, estrus 24-48 hours early.

Features:

- Wearable Sensors: Movement, temperature, activity
- LoRaWAN: 30-minute data intervals
- AI Diagnostics: Early disease detection
- Productivity Tracking: Lactation, reproduction

Expected Outcomes:

- ✓ -30% Veterinary Costs
- ✓ +15% Productivity Gain
- ✓ +8% Milk Quality (SCC improvement)
- ✓ Animal Welfare ↑

FMIS.Equipment - P2P Machinery Marketplace

Overview: Tractor, harvester, sprayer daily/monthly rentals. Geographic matching, smart escrow. Farmers rent instead of buy.

Features:

- Geographic Matching: GPS-based
- Pricing: 5K TL/day (vs. 500K TL purchase)
- Smart Escrow: Payment after renter confirms return
- Insurance: Platform-included

Expected Outcomes:

- ✓ -20% Farmer Equipment Costs
- ✓ +40% Machine Utilization Rate
- ✓ Machine Owner: 200% annual ROI
- ✓ New Business Model: Farmers → Equipment Entrepreneurs

FMIS.AI-Helpdesk - 24/7 Advisory

Overview: 24/7 agricultural advisory. Dual-layer: (1) AI Chatbot (80% questions), (2) Expert escalation (2-hour max). 24-hour wait → 2-hour response.

Features:

- AI Chatbot: 1000+ knowledge base articles
- Turkish Climate: Crop calendars, disease guides
- Expert Escalation: Complex queries → 2 hours
- WhatsApp + Voice: Accessibility

Expected Outcomes:

- ✓ 24-hour → 2-hour advisory access
- ✓ -40% Disease Spread
- ✓ Knowledge Democratization: Every farmer gets expert advice
- ✓ +10% Error Reduction

FMIS.Market-Access - B2B + B2C

Overview: Market access channels. B2B: Direct sales to food companies. B2C: QR + blockchain traceability. +20% price premium.

Features:

- B2B: Ülker, Do■adan direct contracts
- B2C: QR → Farm identity, satellite data, blockchain
- Pricing: Commodity (70%) 25 TL/kg vs. Premium (30%) 30 TL/kg
- Commission: Platform 3% (B2B), 8% (B2C)

Expected Outcomes:

- ✓ Farmer: +10% (B2B), +6% (B2C premium)
- ✓ Food Co: -10% cost, +15% quality
- ✓ Consumer: Farm-to-table transparency
- ✓ Year 3 Revenue: 7.4M TL (B2B+B2C)